Meta-analytic structural equation modeling as a framework for estimating between- and within-individual effects in multilevel dynamic models

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Abstract

TODO

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Substance use during adolescence represents a major public health concern, contributing to morbidity, mortality, and long-term psychosocial difficulties (Patrick et al., 2025). Early onset of substance use is associated with a greater likelihood of dependence, cognitive impairment, and behavioral problems extending into adulthood (Chassin et al., 2009; Squeglia et al., 2009). Youth growing up in **socioeconomically disadvantaged neighborhoods** are particularly vulnerable, as they face heightened exposure to stress, limited access to adaptive coping resources, and greater contextual reinforcement for risk-taking behaviors (Leventhal & Brooks-Gunn, 2000; Wills et al., 2004). Understanding the affective mechanisms that underlie substance-use vulnerability in these high-risk developmental contexts remains a key priority for prevention science.

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Affective processes are central to theories of adolescent substance use. Two complementary motivational models–**negative reinforcement** (substances used to alleviate negative affect; Baker et al., 2004) and **positive reinforcement** (substances used to enhance or prolong positive affect; Cooper et al., 1995)–propose that substance use serves as an emotion-regulation strategy. However, findings linking affect to subsequent substance use are often inconsistent (Hussong et al., 2011; Simons et al., 2010), perhaps because most research has focused on *mean levels* of affect rather than *dynamic processes* that unfold across time. For adolescents, whose affective systems are still maturing and particularly reactive to contextual stressors (Casey et al., 2008), examining the *temporal organization* of affect may reveal how emotion regulation contributes to early risk for substance use.

Emerging research highlights that affect dynamics—how emotions persist and influence one another over time—offer important insights into emotion regulation and psychopathology (Kuppens, Allen, & Sheeber, 2010; Houben, Van Den Noortgate, & Kuppens, 2015). Within-person autoregressive (AR) effects index **emotional inertia**, or the extent to which affective states carry over from one moment to the next (Kuppens et al., 2010). High AR in negative affect (NA) reflects slow recovery from distress, suggesting sustained dysphoria or rumination, whereas high AR in positive affect (PA) reflects the capacity to sustain positive emotional states (Hollenstein, 2015). Cross-regressive (CR) effects, by contrast, index **coupling** between affective systems–how one emotion predicts changes in another over time (Bringmann, Pe, Vissers, Ceulemans, & Kuppens, 2016; Chow, Zu, Shifren, & Zhang, 2011). For instance, a strong negative NA→PA effect indicates that high negative mood suppresses subsequent positive mood, signaling poor emotional recovery, while a strong negative PA→NA effect indicates that positive experiences can buffer against later distress.

From a developmental and contextual perspective, these affective dynamics may be especially consequential for adolescents in low-SES environments, where chronic stressors (e.g., neighborhood violence, financial strain, unstable relationships) challenge emotion regulation systems (Evans & Kim, 2013; Brody et al., 2013). Persistent negative affect and weak cross-affective recovery may increase the likelihood that youth turn to substances as a means of coping–consistent with the **negative reinforcement model**–whereas stronger positive-affect inertia or adaptive PA→NA dampening may confer resilience. In this sense, AR and CR parameters represent *individualized dynamic signatures* of emotion regulation capacity (Hamaker, Asparouhov, Brose, Schmiedek, & Muthén, 2018), which may distinguish adolescents who escalate to substance use from those who do not.

Despite these theoretical links, few studies have examined whether individual differences in affective dynamics—as captured by AR and CR parameters—predict later substance use and behavioral problems among at-risk adolescents. Existing research has largely focused on concurrent affect–use associations or on static variability metrics such as affective instability (Pe et al., 2015). The present study addresses this gap by modeling adolescents' daily positive and negative affect using a multilevel vector autoregressive (VAR) framework, estimating person-specific AR and CR parameters, and testing whether these dynamic features prospectively predict future substance use and externalizing behavior. By focusing on affective dynamics rather than static mood levels, this study aims to clarify how the persistence and coupling of affective states function as early markers of risk and resilience in socioeconomically disadvantaged youth.

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